

Julius Zhang

Ph.D. Student in Computer Science
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EXPERIENCE

Columbia University

Visiting Researcher

Jun. 2026 - Present

- Conduct research in the Department of Mathematics, hosted by Andrew Blumberg.
- Computational optimal transport; arithmetic geometry applied to cryptography; security of AI-automated formal verification; homotopy theory.

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Researcher

Apr. 2025 - Jul. 2026

- Research on new cryptographic constructions from lattice-based cryptography and commitment compression techniques. Manuscripts forthcoming.
- Building [Jolt](#), open-source zero-knowledge virtual machine in Rust. Implemented and benchmarked [torus compression](#), yielding a $3\times$ reduction in commitment size; proved optimality with a conceptual reframing via Hilbert 90. Performance engineering across the proving stack, including sum-check optimization.
- Audited proof systems security: verify correctness of [elliptic-curve optimizations](#) and formalize folklore results into rigorous [proofs](#).
- Evaluated lattice-based cryptography parameters against known attacks under various security regimes and performance constraints.

Millennium

Quantitative Researcher

Jun. 2024 - Apr. 2025

- Convex optimization and robust statistical estimation for real-time trading systems.

Stellar Development Foundation

Consultant

Jun. 2023 - Sept. 2023

- Byzantine fault tolerance, C++, [stellar-core](#).
- Improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.

Distributed Systems and TCS Research, Stanford University

Researcher

Sept. 2023 - Jun. 2024

- Under David Mazières, research in general open membership Byzantine fault tolerance protocols ([Github](#)), with applications to certificate authority management and automatic password recovery with hardware security modules.
- Research PCP under Aviad Rubinfeld, reconstructed proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.

Topology Research, Stanford University

Researcher

Jul. 2020 - May 2022

- Floer homotopy theory under Mohammed Abouzaid; related the complex cobordism Gysin map to the Grothendieck residue symbol and derived a recursive formula for Gromov–Witten quantum product corrections.
- Research under Ciprian Manolescu. Floer theory, Seiberg–Witten, knot invariants. Produced new potential counterexamples for the smooth Poincaré conjecture.

SELECTED PAPERS AND MANUSCRIPTS

- Zachary DeStefano, Noah Golub, Zile Huang, **Julius Zhang**, Sam Frank, and Michael Walfish
Spain: Succinct Proofs for Numerical Computations. [\[PDF\]](#)
USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2026.
- Julius Zhang.
The Relative Trace-Zero Subgroup of the Barreto-Naehrig Curves. [\[PDF\]](#)
Cryptology ePrint Archive, 2026.
- Julius Zhang.
Complex Cobordism and Formal Group Law. [\[PDF\]](#)
Stanford University, 2024.

TALKS

- *p-Ordinary Cohomology for $SL(2)$ over Number Fields.*
Hida Theory Seminar, Columbia University, 2026.
- *Construction of Seiberg-Witten-Floer stable homotopy types.*
Homotopy Theory and Manifolds Seminar, Columbia University, 2026.
- *Bordered Heegaard Floer Homology and the Surgery Exact Triangle.*
Stanford University, 2022.
- *Knots, links, and the search for exotic 4-spheres.*
Stanford University, 2021.

EDUCATION

Courant Institute of Mathematical Sciences, New York University

Ph.D. in Computer Science

June 2026 - Present

Stanford University

M.S. in Computer Science; B.A. in Mathematics

Sept. 2019 - Jun. 2024

General Wang Yaowu Scholarship.

SKILLS

Computer Science. Rust, C/C++, Python; optimization and robust estimation; high-performance numerical systems; cryptography and verifiable computation; formal verification.

Mathematics. Number theory, algebraic geometry, homotopy theory.